

Week 3 — Ubuntu Server Installation Guide

1. Project Overview

ABC Startup Solutions requires its first Linux server to support future enterprise services. The server will eventually be used for file sharing, remote administration, database hosting, web hosting, and internal services.

As a Junior System Administrator, the objective of this project is to deploy a clean Ubuntu Server virtual machine, configure the required settings, enable secure remote administration through SSH, verify that the server is functioning correctly, and document the complete installation procedure.

The installation is performed inside a virtual machine so that the server environment can be safely tested before being deployed to production.

2. Objectives

The objectives of this installation are to:

1. Install Ubuntu Server LTS.
2. Configure the virtual machine according to company requirements.
3. Configure the keyboard and network.
4. Assign the hostname `server01`.
5. Create a non-root administrative account.
6. Configure the storage using the guided installation option.
7. Install OpenSSH Server.
8. Verify the hostname and IP address.
9. Test Internet connectivity.
10. Update the operating system.
11. Verify that the SSH service is running.
12. Document the installation for future administrators.

3. Software Requirements

The following software is required:

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO
- Windows Server Evaluation ISO for the bring-home activity

4. Virtual Machine Specifications

Component	Specification
VM Name	Ubuntu-Server-Week03
Operating System	Ubuntu Server LTS
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB
Disk Format	VDI/VMDK
Network	NAT
Hostname	server01
SSH	OpenSSH Server

5. Installation Procedure

Step 1 — Start the Virtual Machine

Start `Ubuntu-Server-Week03` and boot from the Ubuntu Server LTS ISO.

Screenshot Required: Ubuntu Server installation screen.

Step 2 — Select Language

Select:

English

Continue to the next screen.

Screenshot Required: Language selection.

Step 3 — Configure Keyboard

Select the appropriate keyboard layout.

Recommended:

English (US)

Continue.

Screenshot Required: Keyboard configuration.

Step 4 — Configure Network

Allow Ubuntu Server to obtain an IP address automatically using DHCP unless another configuration is specifically required.

Record the assigned IP address.

Example:

```
IP Address: [YOUR ACTUAL IP ADDRESS]
```

The actual IP address must be taken from the installed virtual machine.

Screenshot Required: Network configuration/IP information.

Step 5 — Configure Hostname

Set the hostname to:

```
server01
```

This hostname will identify the server on the network.

Screenshot Required: Hostname configuration.

Step 6 — Create the Administrative User

Create a non-root administrative account.

Example:

```
Full Name: Ejhay Feliciano  
Username: feliciano  
Password: [YOUR STRONG PASSWORD]
```

The password should not be included in screenshots or documentation.

Screenshot Required: User configuration screen.

Step 7 — Configure Storage

Select:

Guided – Use Entire Disk

Use the entire 40 GB virtual disk.

The installation should automatically create the required Linux partition structure.

Document:

- Disk size: 40 GB
- Partitioning method: Guided – Use Entire Disk
- File system: Ubuntu's default Linux filesystem configuration
- Disk type: VDI/VMDK

Screenshot Required: Storage configuration.

Step 8 — Install OpenSSH Server

Enable:

Install OpenSSH Server

SSH is required because it provides secure remote administration of the server.

Screenshot Required: SSH selection screen.

Step 9 — Additional Packages

Do not install additional packages unless specifically instructed.

Step 10 — Complete Installation

Allow the Ubuntu Server installation to complete.

Restart the virtual machine when prompted.

If prompted to remove the installation ISO, remove/eject it before continuing.

6. Initial Server Configuration

After restarting, log in using the administrative account created during installation.

The first configuration tasks are:

1. Verify the hostname.
2. Verify the IP address.
3. Test Internet connectivity.
4. Update the operating system.
5. Verify the SSH service.

7. Verification Commands

Verify Hostname

Run:

```
hostname
```

Expected result:

```
server01
```

Screenshot Required: Terminal showing the hostname.

Verify IP Address

Run:

```
ip addr
```

Identify the IP address assigned to the network interface.

Screenshot Required: Terminal showing the IP address.

Test Internet Connectivity

Run:

```
ping -c 4 google.com
```

A successful result should show responses from the destination.

Screenshot Required: Successful ping results.

Update the Server

Run:

```
sudo apt update
```

Then:

```
sudo apt upgrade -y
```

These commands update the package information and install available system updates.

Screenshot Required: Update process.

Verify SSH

Run:

```
systemctl status ssh
```

The service should show:

```
active (running)
```

Screenshot Required: SSH service status.

8. Verification Results

Test	Command	Expected Result	Actual Result
Login	Login with created user	Successful login	[Fill after installation]
Hostname	hostname	server01	[Fill actual result]
IP Address	ip addr	IP displayed	[Fill actual result]
Internet	ping -c 4 google.com	Successful replies	[Fill actual result]
Updates	sudo apt update / sudo apt upgrade -y	Updates completed	[Fill actual result]
SSH	systemctl status ssh	Active (running)	[Fill actual result]

9. Installation Result

The Ubuntu Server virtual machine is considered ready for further configuration once the hostname, network connection, operating system updates, and SSH service have been successfully verified.

The screenshots collected during the installation and verification process provide evidence that the required configuration steps were completed.